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Graded Assessment - Primality Testing

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1. The odds that a randomly picked 300 digit number is prime is closest to

1 / 1 point

- ☐ 1 in 100
- ☐ 1 in 1000
- ☒ 1 in 700.
- ☐ 1 in 1 million

Correct
The density of prime numbers near n is about $1/\ln(n)$, $\ln(10^3 \cdot 300)$ is about 691.

2. When testing for primality of N using trial division, which describes the smallest set of trial divisors needed?

1 / 1 point

- ☐ All prime numbers that are less than the square root of N.
- ☐ All integers greater than 1 and no greater than $N/2$.
- ☒ All prime numbers that are no greater than the square root of N.
- ☐ All integers greater than 1 and less than $\sqrt{N}$.

Correct
Correct. The $\sqrt{N}$ is the largest that the smallest factor of N can be, and only primes need to be checked.

3. Currently, the fastest algorithm for factoring large arbitrary numbers is

1 / 1 point

- ☒ the General Number Field Sieve.
- ☐ the multipurpose quadratic sieve.
- ☐ the Sieve of Eratosthenes.
- ☐ square-and-multiply.

Correct
This is generally fastest for numbers over 100 digits.

4. Fermat's Primality Test will always pass, regardless of the chosen base, for some composite numbers known as

1 / 1 point

- ☐ unlucky numbers.
- ☒ Carmichael numbers.
- ☐ Fermat pseudoprimes.
- ☐ such numbers do not exists. All composite numbers will fail the test for at least one choice of the base.

Correct
This is the definition of a Carmichael number.

5. If a^{n-1} is not congruent to 1 (mod n), then 'a' is referred to as

1 / 1 point

- ☐ a Fermat liar to the compositeness of 'n'.
- ☐ a Fermat witness to the primeness of 'n'.
- ☒ a Fermat witness to the compositeness of 'n'.
- ☐ a Fermat liar to the primeness of 'n'.

Correct
Such a result confirms that 'n' is not prime.

6. If a^{n-1} is congruent to 1 (mod n) and 'n' is composite, then 'a' is referred to as

1 / 1 point

- ☐ a Fermat witness to the compositeness of 'n'.
- ☒ a Fermat liar.
- ☐ a Fermat liar to the compositeness of 'n'.
- ☐ a Fermat witness to the primeness of 'n'.

Correct
Such a result would indicate that 'n' might be prime.

7. If N is a prime number, then it can be shown that

1 / 1 point

- ☐ there are numbers other than 1 and -1 that, when squared, are congruent to 1 (mod N)
- ☐ that x^2 is congruent to 1 for all values of x that are relatively prime to N.
- ☒ that there are exactly two square roots of 1, namely 1 and (N-1), (mod N).
- ☐ x^2 is congruent to 1 if x is the multiplicative inverse of N (mod N).

Correct
This is the basis for the Miller-Rabin primality test.

8. Which of the following values for 'x' are liars for the primality of 221 (using Miller-Rabin)?

1 / 1 point

- ☐ 93
- ☒ 174
- ☐ 5
- ☐ 16

Correct
Correct.

9. The number of Miller-Rabin trials that must be performed to achieve a given level of security

1 / 1 point

- ☒ decreases as the size of the prime number sought increases.
- ☐ has an effectively random relationship to the size of the prime number sought.
- ☐ is independent of the size of the prime number sought.
- ☐ increases as the size of the prime number sought increases.

Correct
This is due to the increasing scarcity of strong liars as the size of the number grows.

10. The fraction of bases that are witnesses to the compositeness of N (if it is an odd composite) is

1 / 1 point

- ☐ is usually about 3/4, but may be more or less.
- ☐ is unpredictable and can be zero.
- ☒ at least 3/4.
- ☐ no more than 3/4.

Correct
This allows probabilities bounds to be placed in the likelihood that a composite number will pass several rounds of Miller-Rabin.

